

Classical Mechanics I (Spring 2026): Final Exam Solution

Jun. 13, 2026

[total 25 pts, closed book/cellphone, no calculator, 90 minutes]

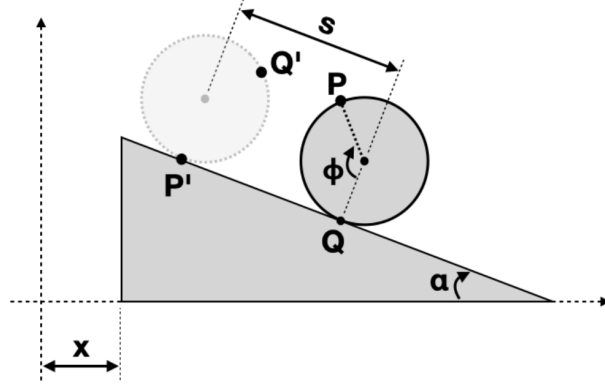
1. [4 pt] Show that the shortest path between two given points on the surface of a right circular cylinder of radius R is a segment of a helix — in three different ways below.
- (a) [2 pt] By introducing a functional in the form $\int f_1\{z, z'\} d\phi$ in cylindrical coordinates $[\rho, \phi, z]$.
- (b) [1 pt] By using an alternative functional form of $\int f_2\{\phi, \phi'\} dz$.
- (c) [1 pt] By rewriting the functional in the form of $\int f_3\{y', z'\} dx$ in Cartesian coordinates $[x, y, z]$, and employing the method of Euler equations with an auxiliary condition imposed.

- (a) $s = \int_1^2 \sqrt{R^2 + z'^2} d\phi \rightarrow z' = \frac{dz}{d\phi} = C_1$
- (b) $s = \int_1^2 \sqrt{R^2 \phi'^2 + 1} dz \rightarrow \frac{R^2 \phi'}{\sqrt{R^2 \phi'^2 + 1}} = C_2 \rightarrow \phi' = \frac{d\phi}{dz} = C_3$.
- (c) $s = \int_1^2 \sqrt{1 + y'^2 + z'^2} dx$ and $g = x^2 + y^2 - R^2 = 0 \rightarrow \frac{d}{dx} \left(\frac{y'}{\sqrt{1 + y'^2 + z'^2}} \right) - 2\lambda(x)y = 0$ and $\frac{d}{dx} \left(\frac{z'}{\sqrt{1 + y'^2 + z'^2}} \right) = 0$. The second equation, $\frac{z'}{\sqrt{1 + y'^2 + z'^2}} = C_4$, combined with the constraint equation, $g = 0$, yields $z'^2 = \frac{C_4^2}{1 - C_4^2} (1 + y'^2) = \frac{C_4^2}{1 - C_4^2} \cdot \frac{R^2}{y^2}$, which essentially means $z' = \frac{C_5}{y}$. Now from $dz = \frac{C_5}{y} dx = \left(\frac{C_5}{R \sin \phi} \right) (-R \sin \phi d\phi) = -C_5 d\phi$, one again finds $z' = \frac{dz}{d\phi} = -C_5$.

2. [4 pt] Consider a homogenous disk of mass m and radius R , which rolls without slipping down the inclined face of a triangular block of mass M . The incline makes an angle α with the horizontal (see figure). The block itself is free to move on a horizontal surface without friction.
- (a) [1 pt] Determine Lagrange's equations of motion for the disk and the block (without using undetermined multipliers). You will first need to clearly define your choice of generalized coordinates. You may include your own diagram on the answer sheet, if it is helpful.
- (b) [1 pt] Determine the acceleration of the disk down the inclined face.

(c) [1 pt] Repeat (a) using the Lagrangian method with undetermined multipliers. Identify the force(s) of constraint and explain their physical meaning(s).

(d) [1 pt] Repeat (a)-(b) using the Newtonian method, in the special case where the block is immovable. Compare the resulting downward acceleration of the disk with your answer in (b) by taking the limit $M \rightarrow \infty$.



- (a) x : distance traveled by the block M , s : distance traveled by the disk m down the incline, then, $T = \frac{3}{4}m\dot{s}^2 + \frac{1}{2}(m+M)\dot{x}^2 + m\dot{x}\dot{s}\cos\alpha$, $U = -mgs\sin\alpha \rightarrow (m+M)\ddot{x} + m\ddot{s}\cos\alpha = 0$, and $m\ddot{x}\cos\alpha + \frac{3}{2}m\ddot{s} - mg\sin\alpha = 0$.

- (b) Solving the coupled equations above, one finds $\ddot{s} = \frac{g\sin\alpha}{\frac{3}{2} - \frac{m}{m+M}\cos^2\alpha}$.

- (c) $T = \frac{1}{2}m\dot{s}^2 + \frac{1}{4}mR^2\dot{\phi}^2 + \frac{1}{2}(m+M)\dot{x}^2 + m\dot{x}\dot{s}\cos\alpha$, $U = -mgs\sin\alpha$, and $g(s, \phi) = s - R\phi = 0 \rightarrow (m+M)\ddot{x} + m\ddot{s}\cos\alpha = 0$, $m\ddot{x}\cos\alpha + m\ddot{s} - mg\sin\alpha + \lambda = 0$, and $\frac{1}{2}mR^2\ddot{\phi} - \lambda R = 0$. From the last equation one can see the physical meaning of the force of constraint — the static friction force between the disk and the incline.

- (d) $m\ddot{s} = mg\sin\alpha - F$, and $FR = \frac{1}{2}mR^2\left(\frac{\ddot{s}}{R}\right) \rightarrow \ddot{s} = \frac{2}{3}g\sin\alpha$.

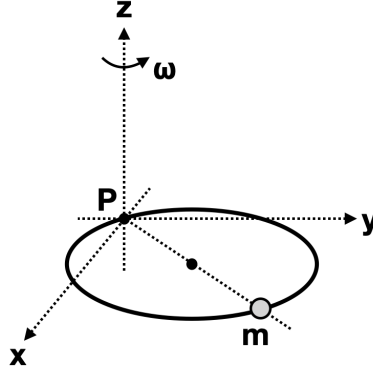
3. [4 pt] Consider a bead of mass m constrained to move without friction on a horizontal circular wire of radius R . The wire lies in its own plane and rotates at a constant angular velocity ω about a fixed point P located on the wire (see figure).

(a) [1 pt] Determine the Lagrangian describing the motion of the bead. You will first need to clearly define your choice of generalized coordinates. You may include your own diagram on the answer sheet, if it is helpful.

(b) [1 pt] Determine Lagrange's equation of motion. Show that the bead's motion, measured with respect to the diameter passing through point P and the center of the wire, is very similar to that of a *plane pendulum* in a uniform gravitational field.

(c) [1 pt] Explain qualitatively why this equivalence is physically reasonable. Then, find the period of small oscillations.

(d) [1 pt] Repeat (b) using the Hamiltonian method.

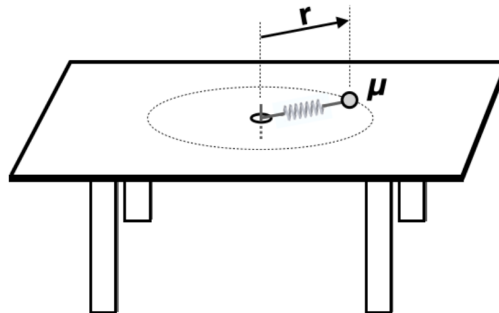


- (a) θ : displacement from the equilibrium point on the “diameter” described in the problem, then, $T = L = \frac{1}{2}m \left(R^2\omega^2 + R^2(\dot{\theta} + \omega)^2 + 2R^2\omega(\dot{\theta} + \omega) \cos \theta \right)$.
- (b) $-mR^2\omega(\ddot{\theta} + \omega) \sin \theta - mR^2(\ddot{\theta} - \omega\dot{\theta} \sin \theta) = -mR^2(\ddot{\theta} + \omega^2 \sin \theta) = 0$.
- (c) The centrifugal force always points away from the pivot point P, and the component of this force along the wire acts as a restoring force just as gravity does along the arc of a plane pendulum. The diameter through P acts like the “vertical” direction of a pendulum.
- (d) From L in (a), $p_\theta = \frac{\partial L}{\partial \dot{\theta}} = mR^2(\dot{\theta} + \omega + \omega \cos \theta) \rightarrow H = p_\theta \dot{\theta} - L = mR^2(\dot{\theta}^2 + \omega\dot{\theta} + \omega\dot{\theta} \cos \theta) - \frac{1}{2}m \left(R^2\omega^2 + R^2(\dot{\theta} + \omega)^2 + 2R^2\omega(\dot{\theta} + \omega) \cos \theta \right) = \frac{1}{2}mR^2\dot{\theta}^2 - mR^2\omega^2(1 + \cos \theta) = \frac{1}{2} \cdot \frac{[p_\theta - mR^2\omega(1 + \cos \theta)]^2}{mR^2} - mR^2\omega^2(1 + \cos \theta) \rightarrow \dot{\theta} = \frac{\partial H}{\partial p_\theta} = \frac{p_\theta - mR^2\omega(1 + \cos \theta)}{mR^2}$ (as expected) and $\dot{p}_\theta = -\frac{\partial H}{\partial \theta} = -\frac{p_\theta - mR^2\omega(1 + \cos \theta)}{mR^2} \cdot mR^2\omega \sin \theta - mR^2\omega^2 \sin \theta = -mR^2(\dot{\theta} + \omega)\omega \sin \theta \rightarrow$ combined, $\dot{p}_\theta = -mR^2(\ddot{\theta} + \omega)\omega \sin \theta = mR^2(\ddot{\theta} - \omega\dot{\theta} \sin \theta) \rightarrow \ddot{\theta} + \omega^2 \sin \theta = 0$.

4. [4 pt] A particle of mass μ is attached to a massless string of spring constant k , the other end of which is fixed to a point on a horizontal table (see figure). The particle moves without friction under the influence of an attractive central force $F(r) = -k(r - a)$ where r is the distance from the particle to the fixed point.

(a) [2 pt] Using the concept of an effective potential, find the condition for radius r_0 at which the particle with angular momentum ℓ moves in a circular orbit.

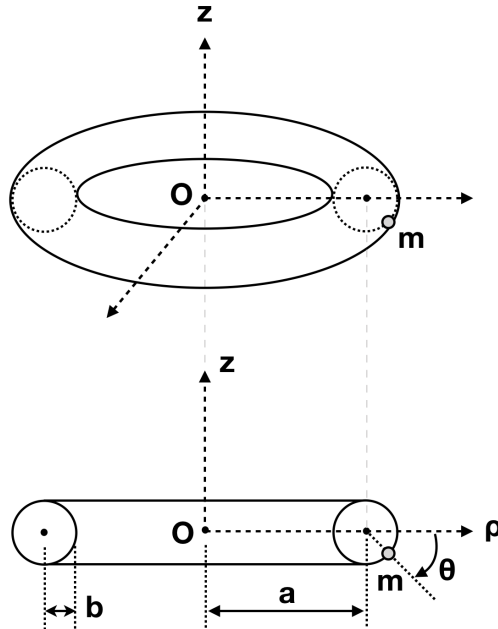
(b) [2 pt] Then, again using the concept of the effective potential, show that a slight radial nudge to this circular orbit causes only small radial oscillations. Find the period for these oscillations.



- (a) $\left. \frac{\partial V(r)}{\partial r} \right|_{r_0} = 0$ with $V(r) = \frac{1}{2}k(r-a)^2 + \frac{\ell^2}{2\mu r^2} \rightarrow k(r_0-a) = \frac{\ell^2}{\mu r_0^3}$.
- (b) For the stability analysis, we use Eq.(8.76): $\left. \frac{\partial^2 V(r)}{\partial r^2} \right|_{r_0} = k + \frac{3\ell^2}{\mu r_0^4} > 0$ indicating that the orbit is stable. Then, $\omega_{\text{osc}} = \sqrt{\frac{[\partial^2 V(r)/\partial r^2]_{r_0}}{\mu}} = \sqrt{\frac{1}{\mu} \left(k + \frac{3\ell^2}{\mu r_0^4} \right)}$, or, $T_{\text{osc}} = 2\pi \sqrt{\frac{\mu r_0}{k(4r_0-3a)}}$.

5. [5 pt] A particle of mass m is constrained to move on the smooth inner surface of a torus, as shown in the figure below. The particle is subject to the gravitational field g along the symmetry axis of the torus ($-z$ direction). The position of the particle is then described by $(\rho, \phi, z) = (a + b \cos \theta, \phi, -b \sin \theta)$, where ϕ is the azimuthal angle in the cylindrical coordinates and θ is the angle measured in the vertical plane containing the particle with respect to the $z = 0$ plane (see figure).

- (a) [1 pt] Determine Lagrange's equation(s) of motion for the particle.



- (b) [1 pt] Find the generalized momenta, p_ϕ and p_θ , and Hamilton's equation(s) of motion.
- (c) [2 pt] Discuss the existence of an angle θ_0 at which θ remains unchanged — i.e., the particle is in the state of uniform circular motion. Find the angular frequency ω_{orb} of this circular motion. (Note: You may find it helpful to introduce an effective potential $V(\theta)$ and analyze the particle's motion for different values of p_ϕ .)
- (d) [1 pt] Show that θ_0 corresponds to a stable equilibrium position. Demonstrate that slight perturbation in θ leads to small oscillations about θ_0 with the angular frequency of

$$\omega_{\text{osc}} = \sqrt{\frac{g}{b \sin \theta_0} + 3 \omega_{\text{orb}}^2 \sin^2 \theta_0}.$$

- (a) $T = \frac{1}{2}m(\dot{\rho}^2 + \rho^2\dot{\phi}^2 + \dot{z}^2) = \frac{1}{2}m[b^2\dot{\theta}^2 + (a + b\cos\theta)^2\dot{\phi}^2]$, $U = -mgb\sin\theta \rightarrow -mb^2\ddot{\theta} - m(a + b\cos\theta)b\sin\theta\dot{\phi}^2 + mgb\cos\theta = 0$, and $\frac{d}{dt}[m(a + b\cos\theta)^2\dot{\phi}] = 0$.
- (b) With $p_\phi = m(a + b\cos\theta)^2\dot{\phi}$ and $p_\theta = mb^2\dot{\theta}$, $H = \frac{p_\theta^2}{2mb^2} + \frac{p_\phi^2}{2m(a+b\cos\theta)^2} - mgb\sin\theta \rightarrow \dot{p}_\phi = -\frac{\partial H}{\partial \phi} = 0 = \frac{d}{dt}[m(a + b\cos\theta)^2\dot{\phi}]$, and $\dot{p}_\theta = -\frac{\partial H}{\partial \theta} = -m(a + b\cos\theta)b\sin\theta\dot{\phi}^2 + mgb\cos\theta = mb^2\ddot{\theta}$, which are consistent with (a).
- (c) $H = E_{\text{tot}} = \frac{p_\theta^2}{2mb^2} + \frac{p_\phi^2}{2m(a+b\cos\theta)^2} - mgb\sin\theta = \frac{p_\theta^2}{2mb^2} + V(\theta) \rightarrow \left.\frac{\partial V(\theta)}{\partial \theta}\right|_{\theta_0} = 0 = \frac{p_\phi^2 \cdot b \sin\theta_0}{m(a+b\cos\theta_0)^3} - mgb\cos\theta_0 = m(a + b\cos\theta_0)b\dot{\phi}^2\sin\theta_0 - mgb\cos\theta_0$. One can then find $\omega_{\text{orb}} = \dot{\phi} = \sqrt{\frac{g\cos\theta_0}{(a+b\cos\theta_0)\sin\theta_0}} = \sqrt{\frac{g}{a\tan\theta_0 + b\sin\theta_0}}$. Alternatively, you can acquire the same result by imposing $\ddot{\theta} = 0$ in the equation for θ in (a) or (b).
- (d) $\left.\frac{\partial^2 V(\theta)}{\partial \theta^2}\right|_{\theta_0} = \frac{p_\phi^2 \cdot b \cos\theta_0}{m(a+b\cos\theta_0)^3} + \frac{3p_\phi^2 \cdot b^2 \sin^2\theta_0}{m(a+b\cos\theta_0)^4} + mgb\sin\theta_0 = m(a+b\cos\theta_0)b\dot{\phi}^2\cos\theta_0 + 3mb^2\dot{\phi}^2\sin^2\theta_0 + mgb\sin\theta_0 = mgb\cos\theta_0 \left(\frac{\cos\theta_0}{\sin\theta_0}\right) + 3mb^2\dot{\phi}^2\sin^2\theta_0 + mgb\sin\theta_0 = \frac{mgb}{\sin\theta_0} + 3mb^2\dot{\phi}^2\sin^2\theta_0 \rightarrow \text{Thus, } \omega_{\text{osc}} = \sqrt{\frac{1}{mb^2} \cdot \left.\frac{\partial^2 V(\theta)}{\partial \theta^2}\right|_{\theta_0}} = \sqrt{\frac{g}{b\sin\theta_0} + 3\omega_{\text{orb}}^2 \sin^2\theta_0}$.

6. [4 pt] Throughout the semester we discussed many examples in which concepts in classical mechanics are utilized in contemporary research and in explaining daily phenomena.

(a) [2 pt] We discussed how invisible objects in the Universe were discovered by modeling a two-body system interacting via the central force. Describe three or more such cases. 2-3 sentences per case are expected to clearly explain how people came to notice the existence (and mass) of the invisible object using the concept of two-body motion, central force, and/or reduced mass. Use diagrams if desired.

(b) [1 pt] In the last class of the semester, five of your peers presented their term projects. Describe the key idea of one of the presentations you found interesting. A paragraph of at least 3-4 sentences is expected to clearly convey the core idea of his/her term project. If you were one of the presenters, please choose someone else's.

(c) [1 pt] We also discussed how one can speedily gain insights into a physical phenomenon, by using techniques such as order-of-magnitude estimation and/or dimensional analysis. Invent and solve your own order-of-magnitude estimation problem. Start with a paragraph of at least 2-3 sentences to clearly describe the problem setup. Make a physically intuitive, yet simple problem so that you can explain your problem *and* solution to a fellow physics major student in ~ 3 minutes. Use diagrams if desired. Do not plagiarize another person's idea.

- (a) Exoplanet systems, Pluto+Charon system, Sirius A+B system, Cygnus X-1 system, supermassive black hole and nearby stars. For more information about each of the items above, see the class slides, Lectures 1-1 and 14-1.

- (b) See the student presentation slides in Lecture 15-1 that include the collection of term project presentations by students on June 9.

- (c) See the class slides for Lecture 13-1 that include many example problems, and the grading guideline.